

Dynamical crossover at $T^* \simeq 0.55 T_{\text{BKT}}$ in an hBN-encapsulated $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+\delta}$ weak link, consistent with self-heating-limited switching

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We analyze the zero-field current-driven transport of a single hBN-encapsulated exfoliated $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+\delta}$ (Bi-2212) weak link at 27 temperatures between 2 and 110 K. Resistance-temperature curves at four bias currents (0.316–10 μA) collapse onto a Halperin–Nelson tail giving $T_{\text{BKT}} = 86.7$ K, with a fit-window systematic of ± 0.3 K and a bias spread of ± 0.2 K. Three independently constructed observables—the supercurrent collapse rate $|dI_c/dT|$, the switching–retrapping hysteresis $1 - I_r/I_c$, and the supercurrent nonreciprocity $|\eta|$ —peak at 45–50 K. A joint bootstrap over repeated sweeps reproduces each peak location in every resample, so the three maxima coincide within one step of the 5-K temperature grid and define a crossover at $T^* = 45\text{--}50$ K, i.e. $T^*/T_{\text{BKT}} \simeq 0.55$. The hysteresis minimum $I_r/I_c = 0.74$ at 50 K is stable in position for voltage criteria spanning 0.3–2 mV, and a polarity-symmetric/antisymmetric decomposition—validated on synthetic data—recovers its intrinsic component without bias while bounding inter-sweep thermal drift at $\lesssim 0.3$ K. Principal-component analysis of threshold-free I – V lineshape descriptors places all temperatures on a one-dimensional manifold whose leading coordinate reorganizes over the same intermediate range; a leave-one-descriptor-out audit shows, however, that a single change point of this coordinate moves between 27.5 and 77.5 K, so we do not use it to locate T^* . An effective thermal conductance $G \sim 0.01\text{--}0.13$ $\mu\text{W/K}$ extracted from the measured retrapping points, inserted into a hot-spot power-balance model, reproduces the non-monotonic $I_r/I_c(T)$ as an in-sample consistency check, whereas a capacitive (RCSJ) origin would require an unphysical junction capacitance of ~ 0.04 aF. The observations are consistent with self-heating-limited (hot-spot) switching crossing over to overdamped flux flow at T^* . We present the analysis—concordance of independently constructed observables, each audited against the analysis choices behind it—as a protocol for the single-device limit common to air-sensitive two-dimensional superconductors.

I. INTRODUCTION

Atomically thin crystals of the high- T_c cuprate $\text{Bi}_2\text{Sr}_2\text{CaCu}_2\text{O}_{8+\delta}$ (Bi-2212) retain superconductivity down to the monolayer limit [1], making them a natural platform for studying two-dimensional (2D) superconductivity, where the resistive transition is governed by Berezinskii–Kosterlitz–Thouless (BKT) vortex physics [2, 3], and, more recently, for nonreciprocal supercurrent phenomena [4, 5]. Because few-unit-cell Bi-2212 degrades within minutes in ambient conditions, such experiments are typically restricted to a small number of devices—often a single one—so the central experimental question becomes: *what can be established rigorously from one device?* Claims that rest on a single extracted number (a critical temperature, a diode efficiency) are vulnerable to threshold choices and measurement artifacts. Here we take a different route: we extract several independently constructed observables from a dense temperature series of current–voltage (I – V) characteristics, quantify the uncertainty of each by resampling, quantify their residual statistical correlations, audit each against the analysis choices that produced it, and ask whether they agree.

Three observables—the supercurrent collapse rate, the switching–retrapping hysteresis, and the supercurrent nonreciprocity—peak in the same band, $T^* = 45\text{--}50$ K $\simeq 0.55 T_{\text{BKT}}$, far below the resistive transition at $T_{\text{BKT}} =$

86.7 K. An unsupervised, threshold-free analysis of the I – V lineshape shows that the curves reorganize along a single latent coordinate over the same temperature range; an audit of the descriptor set, however, shows that this coordinate alone does not pin down T^* , and we report both results. The phenomenology on both sides of T^* , in particular the non-monotonic temperature dependence of the retrapping-to-switching ratio I_r/I_c , is consistent with hysteresis governed by self-heating (hot-spot formation) rather than junction capacitance [6–9]: a power-balance model calibrated on the measured retrapping points reproduces $I_r/I_c(T)$ including its 50-K minimum as an in-sample consistency check (Appendix C), while a capacitive origin fails by more than three orders of magnitude. We phrase the interpretation as “consistent with” throughout: a direct thermal measurement (pulsed I – V or on-chip thermometry) remains the decisive test.

II. DEVICE AND MEASUREMENTS

The device is a mechanically exfoliated Bi-2212 flake transferred onto a $\text{SiO}_2(285\text{ nm})/\text{Si}$ substrate, contacted by four Au electrodes and capped by an hBN flake that protects the crystal from ambient degradation. Two families of zero-field measurements are analyzed: (i) four-terminal I – V sweeps at 28 temperatures between 2 and 120 K, with the current ramped between ± 0.1 mA in

1- μA steps under a ± 0.4 V compliance, in both ramp directions (from $+I_{\text{max}}$ to $-I_{\text{max}}$ and back), with 2–7 sweeps per temperature (14 at 10 K, including the slow control sweeps below); the single 120-K sweep does not survive the upstream cleaning, leaving 131 sweeps at 27 temperatures between 2 and 110 K; and (ii) resistance–temperature $R(T)$ curves through the transition at fixed bias currents of 0.316, 1, 3.16, and 10 μA . Control sweeps at 10 K taken ~ 30 times slower (2 s per point) reproduce the standard-rate switching current to within 3%, bounding sweep-rate effects at that level; repeated sweeps at fixed temperature reproduce I_c with a median scan-to-scan spread of 0.05%, demonstrating the stability of the hBN-protected device over the full campaign.

All analysis is produced by a deterministic, configuration-driven pipeline (Appendix A). Raw sweeps are split into rising- and falling- $|I|$ legs; because the ramp direction and current sign fix whether $|I|$ is increasing or decreasing, each measured point belongs unambiguously to one of four branches: switching ($|I|$ rising) or retrapping ($|I|$ falling), at positive or negative polarity. With this protocol the rising leg at positive polarity and the rising leg at negative polarity occur in *different* sweep files; all cross-file comparisons are therefore explicitly drift-bounded below. Quality control adds boolean flags (voltage rail, noise floor $\sim 1.5 \times 10^{-4}$ V) but never deletes or modifies data. Uncertainties on derived quantities are nonparametric bootstrap intervals over repeated sweeps; quantities that depend on an analysis choice (voltage criterion, fit window, descriptor set) are additionally recomputed across that choice, and the resulting spread is reported as a systematic.

III. RESULTS

A. Transition temperature and its systematics

Figure 1(c) shows the resistive transition at four bias currents spanning a factor of 30. The rising tail is described by the Halperin–Nelson (HN) form [3]

$$R(T) = R_0 \exp\left[-b(T - T_{\text{BKT}})^{-1/2}\right], \quad (1)$$

with $b = 2.5\text{--}3.0$. Statistical uncertainties on T_{BKT} are below 0.03 K; the relevant uncertainties are systematic, and we quantify them in Appendix B: varying the fit window over [85.5–87, 93–97] K moves T_{BKT} within 86.4–87.0 K, the spread across bias currents is 0.2 K, the fitted b and T_{BKT} are strongly anticorrelated ($r = -0.91$), and the residuals show a coherent $\pm(0.02\text{--}0.04)$ Ω structure common to all four currents, indicating a small departure from the pure HN form as expected for an inhomogeneity-broadened transition [13]. We therefore quote $T_{\text{BKT}} = 86.7 \pm 0.3$ K (systematic). The complementary I – V criterion $V \propto I^{\alpha(T)}$, $\alpha(T_{\text{BKT}}) = 3$, is not robust in this device: the accessible dissipative window is truncated from below by the noise floor and from above

by the switching instability, and the apparent $\alpha = 3$ crossings move or vanish as the fit window is varied. We rely on the HN tail, which is threshold-free and reproducible across bias.

B. Threshold-free lineshape analysis

Figure 1(a) shows why any fixed voltage criterion is dangerous here: at 5 K the switching branch jumps discontinuously to hundreds of mV; at 35–55 K a gradual low-voltage foot precedes the jump; near T_{BKT} the response is smooth. A fixed millivolt-scale threshold therefore samples *different physical features* at different temperatures.

To characterize the evolution without such a criterion we describe each rising- $|I|$ branch by six shape descriptors—defined precisely in Appendix A—none of which uses a voltage threshold. Principal-component analysis of the standardized descriptors [Fig. 2(a)] places the 27 temperatures on a single ordered arc (PC1 and PC2 carry 43% and 36% of the variance), so the I – V lineshape evolution is governed by essentially one latent coordinate. We deliberately do *not* cluster these points: on a one-dimensional manifold, hard clustering would impose sharp boundaries on what the data show to be a continuous evolution.

The leading coordinate PC1 rises steeply between ~ 20 and ~ 55 K and shows a second feature near T_{BKT} [Fig. 2(b)]. With all six descriptors, a least-squares single-change-point fit to PC1(T) gives 47.5 K, with a 68% bootstrap interval of 37.5–47.5 K over 300 resamples of the repeated sweeps. That interval reflects sampling noise only. Repeating the analysis with each descriptor removed in turn, or with only the three dimensionless descriptors, moves the change point between 27.5 and 77.5 K [Fig. 2(c)], although seven of the eight estimates have 68% bootstrap intervals no wider than 10 K. The single-change-point model is misspecified for a ramp followed by a second feature, and small changes in the descriptor weights decide which feature it selects; for the eighth set (without descriptor 3) the choice is close enough that the bootstrap itself alternates between the two features. We therefore use the PCA for what it shows robustly—that the lineshape evolves along one coordinate and reorganizes at intermediate temperature—and locate T^* with the current-based observables of Secs. III C and III D.

C. Switching, retrapping, and intrinsic hysteresis

Figure 3(a) shows the switching current I_c and retrapping current I_r (voltage criterion $V_{\text{th}} = 1$ mV, averaged over polarities and repeated sweeps). From $I_c = 64.0$ μA at 2 K, the supercurrent collapses steeply between 35 and 55 K, passes through a shoulder at 55–75 K near 10 μA , and vanishes at T_{BKT} . The ratio I_r/I_c [Fig. 3(b)] is flat

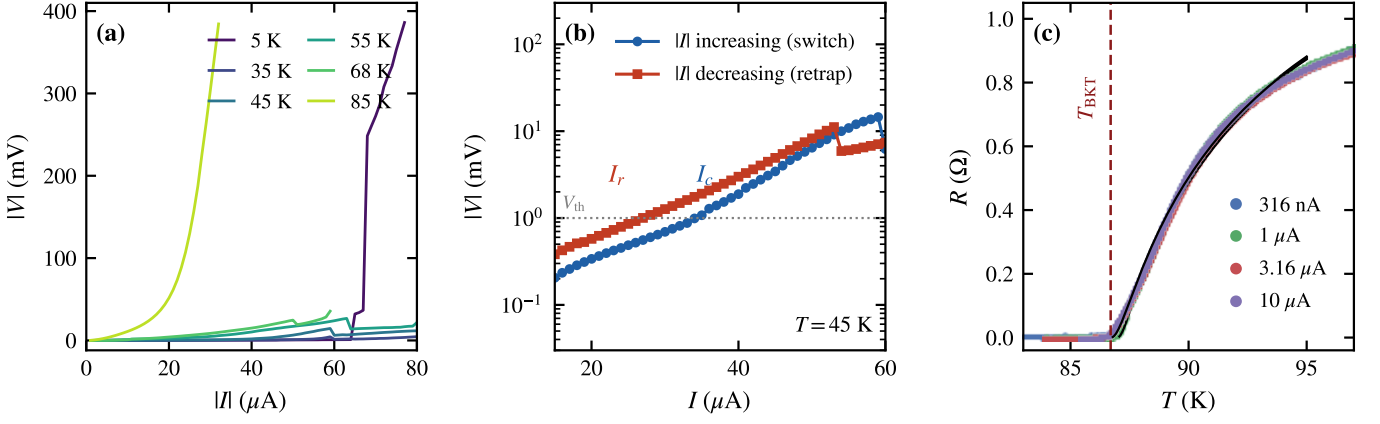


FIG. 1. Transport phenomenology of the hBN-capped Bi-2212 weak link. (a) Rising- $|I|$ (switching) branches of the I - V characteristic at representative temperatures. Three behaviors are visible: an abrupt switch to the dissipative state at low T (5 K), a gradual low-voltage “flux-flow foot” at intermediate T (35–55 K), and a smooth, nearly ohmic response near the transition (85 K). (b) Switching (blue) versus retrapping (red) branches at 45 K on a logarithmic voltage scale; the two branches cross the $V_{\text{th}} = 1$ mV criterion (dotted) at different currents, I_c and I_r , revealing pronounced hysteresis. (c) $R(T)$ at four bias currents with Halperin–Nelson fits [Eq. (1), black]; the fits give $T_{\text{BKT}} = 86.7$ K with a fit-window systematic of ± 0.3 K (dashed line; Appendix B).

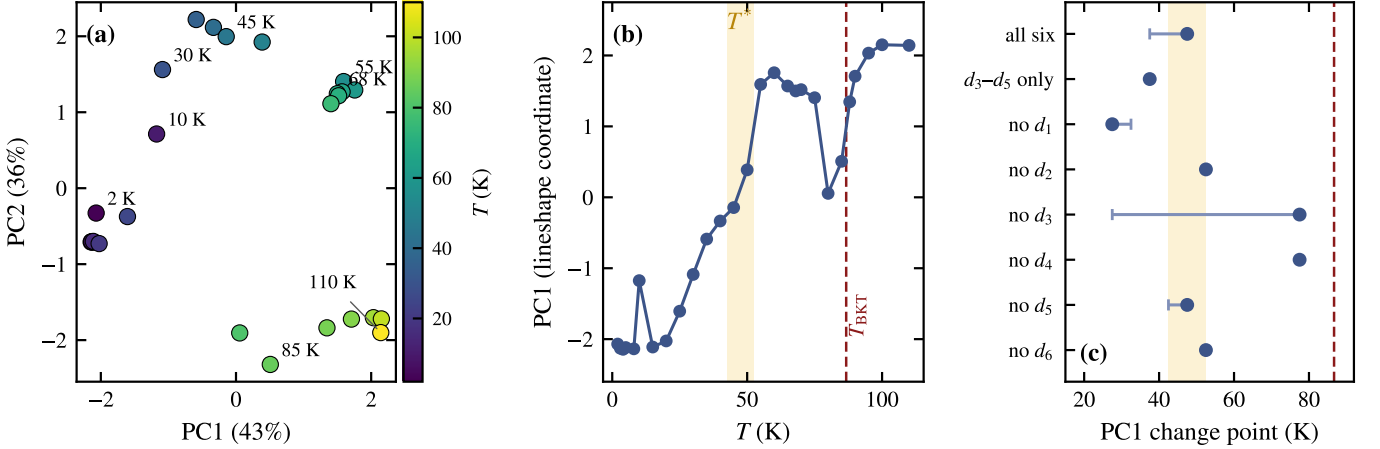


FIG. 2. Threshold-free lineshape analysis. (a) Principal-component scores of the six I - V shape descriptors (Appendix A); each point is one temperature. (b) The leading coordinate PC1 versus temperature. The shaded band marks $T^* = 45$ –50 K as located by the current-based observables (widened by half the 5-K grid step); the dashed line marks T_{BKT} . (c) Change point of PC1(T) for eight descriptor sets—all six, the dimensionless descriptors d_3 – d_5 only, and each descriptor removed in turn—with 68% bootstrap intervals over repeated sweeps. The spread across descriptor sets (27.5–77.5 K) is much wider than seven of the eight bootstrap intervals; without d_3 the bootstrap itself alternates between the two features of PC1(T).

at 0.93 below 30 K, drops to a minimum of 0.74 at 50 K, and recovers to $\simeq 1$ above 60 K. The position of this minimum is not an artifact of the voltage criterion: for V_{th} between 0.3 and 2 mV—almost a decade—the minimum stays at 50–55 K, i.e. within one step of the temperature grid [Appendix B, Fig. 5(a)]. (At $V_{\text{th}} = 5$ mV the criterion enters the flux-flow foot and no longer measures the switching instability; this pathology is documented in Appendix B.)

Because I_c and I_r at a given polarity are extracted from sweeps taken in opposite ramp directions (Sec. II), an apparent hysteresis could be produced by a tempera-

ture offset δT between the two sweep files, of magnitude $\Delta I \approx |dI_c/dT| \delta T$. We separate the two by symmetry: a switching–retrapping hysteresis is *even* under current-polarity reversal, whereas a drift artifact is *odd*. Writing $h_{\pm} = I_c^{\pm} - I_r^{\pm}$,

$$h_{\text{sym}} = \frac{1}{2}(h_+ + h_-), \quad h_{\text{anti}} = \frac{1}{2}(h_+ - h_-). \quad (2)$$

As shown in Fig. 4(a), h_{sym} carries 92% of the total and peaks at 7.0 μ A at 45 K, while h_{anti} is small and uncorrelated with $|dI_c/dT|$ ($r = -0.03$); interpreted as drift it corresponds to $\delta T \lesssim 0.3$ K. We validated the decomposition on synthetic data [Appendix B, Fig. 5(c)]: with a

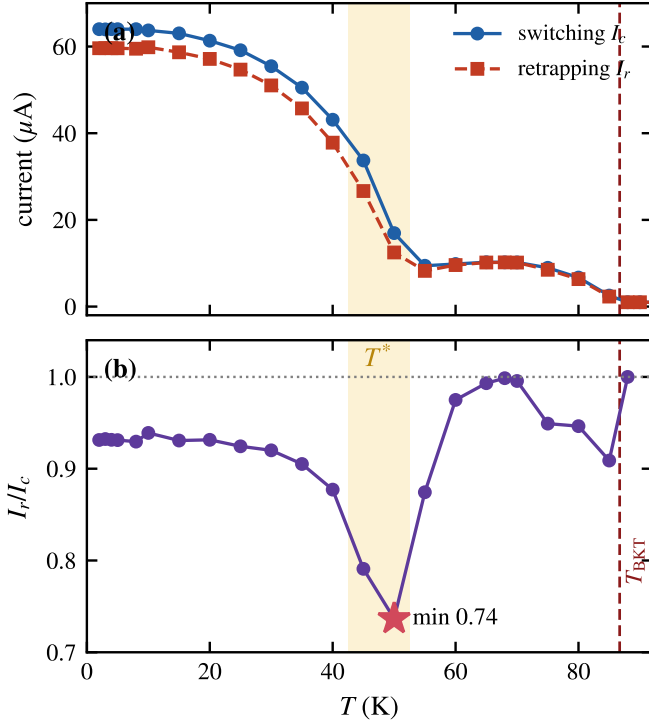


FIG. 3. (a) Switching current I_c and retrapping current I_r versus temperature ($V_{\text{th}} = 1$ mV; polarity- and sweep-averaged). (b) The ratio I_r/I_c : hysteresis is maximal ($I_r/I_c = 0.74$) at 50 K and vanishes above 60 K. Shading: $T^* = 45\text{--}50$ K, widened by half the 5-K grid step; dashed line: T_{BKT} .

known injected $\delta T = 0.3$ K the estimator recovers 0.30 K and the intrinsic component h_{sym} is unbiased to $< 0.2\%$; when an adversarial intrinsic *polarity asymmetry* of the hysteresis is added, h_{sym} remains unbiased but h_{anti} absorbs the asymmetry. Accordingly, we claim only that (i) the intrinsic hysteresis is measured without bias, and (ii) h_{anti} bounds the *combined* magnitude of thermal drift and any intrinsic polarity asymmetry at the equivalent of 0.3 K—not that drift is “fully excluded.”

The supercurrent nonreciprocity $\eta = (I_c^+ - |I_c^-|)/(I_c^+ + |I_c^-|)$ on the switching branch also compares quantities from different sweep files, and inherits the same drift bound: at 50 K, $|dI_c/dT| \delta T / (2I_c) \lesssim 1.5\%$. The measured value rises from $|\eta| < 0.5\%$ below 40 K to 6.2% (bootstrap 95% interval 4.2–8.2%) at 50 K, so at least $\sim 3/4$ of the peak value is not attributable to drift. Without magnetic-field-reversal data we make no claim about its microscopic origin—in particular no superconducting-diode claim [4, 5]; we report it as a bounded observable that peaks in the T^* band.

D. Concordance at T^*

Figure 4(b) overlays, on a common normalized scale, the three current-based observables $|dI_c/dT|$, $1 - I_r/I_c$ and $|\eta|$, together with the six-descriptor lineshape change

rate $|d\text{PC1}/dT|$ for reference. A *joint* bootstrap—resampling the repeated sweeps once per iteration and re-deriving all curves and their extrema simultaneously (400 resamples)—gives peak locations of 45 K ($|dI_c/dT|$) and 50 K ($1 - I_r/I_c$ and $|\eta|$) in every single resample: the statistical uncertainty of each peak position is below the 5-K spacing of the temperature grid, and the appropriate uncertainty to quote is the grid half-spacing, ± 2.5 K. The three maxima therefore coincide within one grid step, and we take $T^* = 45\text{--}50$ K, i.e. $T^*/T_{\text{BKT}} \simeq 0.55$. The lineshape rate also peaks at 50 K, but because the lineshape change point depends on the descriptor set (Sec. III B) we do not count it as independent confirmation.

These observables are *independently constructed*, but not statistically independent, since they share the underlying switching currents to different degrees. We quantify this in Appendix B: the curve-level Pearson correlation over 2–88 K is $r = 0.89$ for $(|dI_c/dT|, 1 - I_r/I_c)$, which share I_c , and $r = 0.35$ and 0.48 for the two pairs involving $|\eta|$. The nonreciprocity is the least correlated of the three and is built from a different combination of branches (the two switching polarities, rather than switching versus retrapping). (The lineshape rate shows a secondary peak at T_{BKT} , where the I – V curves reorganize again at the resistive transition, as expected.)

IV. DISCUSSION

A. A thermal origin of the crossover

The non-monotonic hysteresis is the most discriminant clue to the physics at T^* . In the resistively-and-capacitively-shunted-junction (RCSJ) picture [10, 11], hysteresis requires an underdamped phase and decreases monotonically with temperature. Quantitatively, RCSJ fails here on three counts (Appendix C): with the measured dissipative-branch resistance $R_d \simeq 15$ k Ω and any realistic geometric capacitance (0.1–1 pF), the predicted ratio is $I_r/I_c \sim (2\text{--}6) \times 10^{-4}$, three orders of magnitude below the observed 0.74–0.93; reproducing the observed ratio would require $C \simeq 0.04$ aF; and the predicted temperature dependence is monotonic. Thermally activated escape [12] is likewise negligible: $E_J/k_B = \hbar I_c / 2ek_B \approx 1.5 \times 10^3$ K at 2 K and $\approx 4 \times 10^2$ K at 50 K, so $k_B T / E_J \lesssim 0.12$ throughout.

By contrast, a self-heating description [6–9] is consistent with the data. At each temperature the measured retrapping point (I_r, V_r) defines, through the power balance $P = I_r V_r = G(T_j - T_{\text{bath}})$ with T_j fixed by $I_r = I_c(T_j)$, an effective thermal conductance G between the dissipative region and the bath. The implied G falls in the range 0.08–0.13 $\mu\text{W/K}$ where the device retraps from the fully resistive branch ($T \leq 15$ K, hot-spot excursions $T_j - T_{\text{bath}} \approx 10\text{--}20$ K) and ~ 0.01 $\mu\text{W/K}$ where it retraps from the mV-scale flux-flow state (20–50 K)—both plausible magnitudes for a micrometer-scale flake anchored through ~ 300 nm of SiO_2 , though the factor-

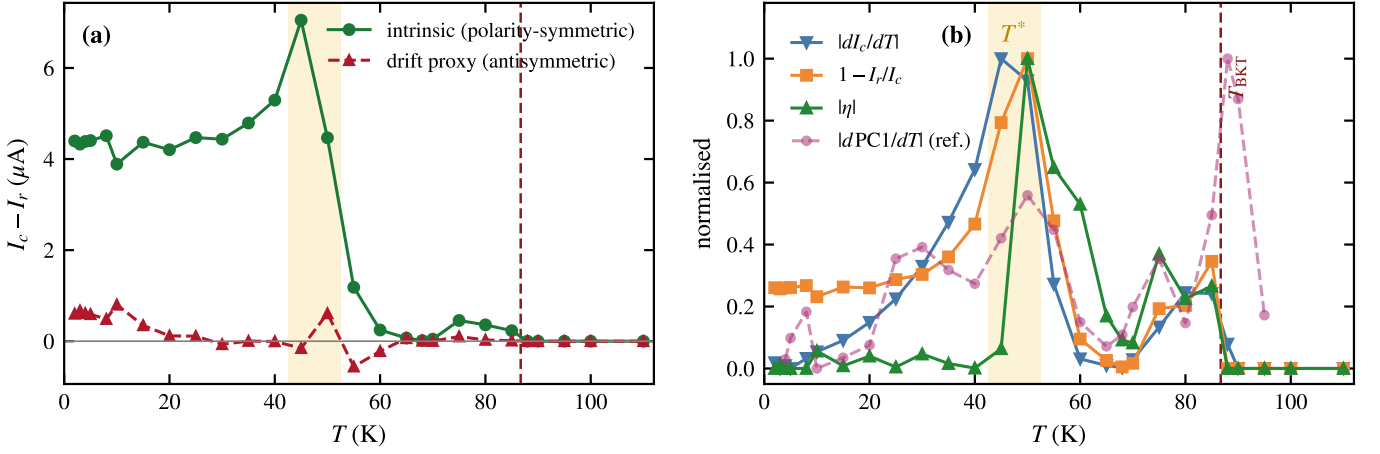


FIG. 4. (a) Polarity-symmetric (intrinsic) and antisymmetric (drift-proxy) components of the switching–retrapping hysteresis $I_c - I_r$ [Eq. (2)]. The symmetric component carries 92% of the signal and peaks at $7.0 \mu\text{A}$ at 45 K ; the antisymmetric component corresponds to $\delta T \lesssim 0.3 \text{ K}$. (b) Concordance of the three current-based observables (each normalized to $[0, 1]$): $|dI_c/dT|$, $1 - I_r/I_c$, $|\eta|$; the six-descriptor lineshape change rate $|dPC1/dT|$ (dashed) is shown for reference only. The three peak in the T^* band (shading); each peak position is reproducible to below the 5-K grid spacing under a joint bootstrap (Appendix B).

of-ten split shows that a single- G hot-spot model is an oversimplification. Inserting a smooth two-parameter interpolation of the calibrated $G(T)$ back into the retrapping condition reproduces $I_r/I_c(T)$, including the 50-K minimum [Appendix C, Fig. 6(b)]. Because G is calibrated on the same retrapping points, this is an in-sample consistency check rather than an independent prediction: it shows that one smooth $G(T)$ is compatible with the whole curve, not that the model is correct. In this picture the regimes separated by T^* acquire a simple meaning: below T^* , switching drives the bridge into a self-sustaining hot spot (voltages jump to the 100-mV scale); around T^* the collapsing supercurrent only marginally sustains the hot spot—here hysteresis and nonreciprocity are maximal and the flux-flow foot emerges; above T^* , dissipation sets in gradually through vortex flow [14] and the dynamics are reversible up to T_{BKT} . We emphasize that this identification is *consistency*, not proof: pulsed I - V protocols or on-chip thermometry would provide the direct thermal evidence.

B. Role of the hBN encapsulation

Encapsulation quality is a live issue for Bi-2212 devices, whose surface degrades by oxygen out-diffusion and water adsorption. Three properties of the present hBN-capped device bear directly on this: (i) $T_{\text{BKT}} = 86.7 \pm 0.3 \text{ K}$ with a sharp HN tail, compared with zero-resistance temperatures of 75.5 – 86.4 K reported for uncapped flakes of comparable and larger thickness on plasma-treated SiO_2 [5]; (ii) a median scan-to-scan reproducibility of I_c of 0.05% across 131 sweeps spanning the full campaign, with no monotonic aging trend; and (iii) an inter-sweep temperature-equivalent stabil-

ity bound of 0.3 K from the antisymmetric hysteresis channel. A single device cannot separate encapsulation from flake-to-flake variation, but these numbers set a useful stability benchmark for hBN capping against which other passivation schemes (uncapped, polymer-capped, or oxide-capped) can be compared on equal statistical footing using the released pipeline.

V. CONCLUSION

From a dense zero-field transport dataset on one hBN-encapsulated Bi-2212 weak link we have shown that (i) the resistive transition sits at $T_{\text{BKT}} = 86.7 \pm 0.3 \text{ K}$ (systematic), reproducible across a 30-fold range of bias current, with fully disclosed fit-window systematics; (ii) three independently constructed observables—the supercurrent collapse rate, the switching–retrapping hysteresis and the nonreciprocity—peak together at $T^* = 45$ – $50 \text{ K} \simeq 0.55 T_{\text{BKT}}$, each reproducible to below the temperature grid under a joint bootstrap; (iii) the hysteresis is intrinsic—its symmetric component is bias-free under a validated decomposition that also bounds drift at 0.3 K —and maximal at T^* ($I_r/I_c = 0.74$, criterion-stable from 0.3 to 2 mV), where the bounded nonreciprocity (6.2% , of which $\lesssim 1.5\%$ could be instrumental) also peaks; (iv) a threshold-free principal-component analysis shows that the I - V lineshape evolves along a single coordinate, but a descriptor audit shows that its change point is not a reliable estimator of T^* in this dataset; and (v) a power-balance model calibrated on the measured retrapping points reproduces the non-monotonic $I_r/I_c(T)$ in sample, while capacitive and fluctuation mechanisms fail quantitatively by orders of magnitude. Beyond the specific device, the methodology—concordance of inde-

pendently constructed, resampled observables, each audited against the analysis choices behind it, in place of any single thresholded number—offers a template for defensible transport claims in the single-device limit.

DATA AND CODE AVAILABILITY

The processed data, configuration files, analysis pipeline, and figure-generation code are available at <https://github.com/LaipoHsun/BSCCO-superconducting-2D-material-analysis->. Every figure in this paper is regenerated by the scripts in `pipeline/` from the cleaned I - V and R - T data included in the repository; the upstream cleaning steps are documented there. The raw instrument files are not public for privacy reasons and are available from the authors upon reasonable request.

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Appendix A: Analysis definitions

Critical currents. For each branch (switching/retrapping $\times$ polarity) of each sweep, I_c (or I_r) is the current at which $|V|$ first crosses $V_{th} = 1$ mV, obtained by linear interpolation between the bracketing samples on the $|I|$ -sorted branch. Aggregates are means over polarities and repeated sweeps; intervals are percentile bootstrap over sweeps (2000 resamples unless stated).

Lineshape descriptors. Each rising- $|I|$ branch with at least eight samples and maximum voltage V_{max} above the noise floor $\sigma_V \simeq 1.5 \times 10^{-4}$ V is reduced to: (1) I_{50} , the current at which $|V|$ first reaches $0.5 V_{max}$ (interpolated); (2) V_{max}/V_{comp} with $V_{comp} = 0.4$ V the compliance; (3) the switching width $(I_{90} - I_{10})/I_{50}$, where I_f is the current at $f V_{max}$; (4) the dissipationless fraction of samples with $|V| < 5\sigma_V$; (5) the foot fraction of samples with $1 \text{ mV} < |V| < 0.1 V_{max}$; and (6) the slope dV/dI over the top quartile of the branch in $|I|$. Descriptors (3)–(5) are dimensionless; (1) is a current, (2) is normalized to the fixed compliance, and (6) is a resistance, so the PCA retains current-scale information through (1). With all six descriptors the PC1 loadings are -0.43 , -0.21 , $+0.58$, -0.60 , $+0.15$ and -0.24 for (1)–(6).

PCA and change point. Descriptors are averaged over sweeps at each temperature, standardized to zero mean and unit variance across temperatures, and decomposed by singular-value decomposition; PC1 is sign-oriented to increase with T . The change point of $PC1(T)$ minimizes the total within-segment sum of squares over all binary partitions of the T -ordered sequence; its statistical uncertainty is the 16–84% interval over 300 bootstrap

resamples of the per-sweep descriptor sets, each resample repeating the standardization, PCA and change-point search. The descriptor audit repeats the whole procedure for eight descriptor sets: all six, (3)–(5) only, and each descriptor removed in turn.

Derivatives. dI_c/dT and $dPC1/dT$ use second-order central differences on the non-uniform temperature grid (`numpy.gradient`); PC1 is first smoothed with a 3-point boxcar, and the two points at each end of the grid, where the boxcar and difference stencils are one-sided, are excluded from peak searches. Peak searches are restricted to $T \leq 80$ K, below the reorganization associated with T_{BKT} itself.

Appendix B: Robustness and systematics

Threshold sensitivity (Fig. 5(a)). The I_r/I_c minimum sits at 50 K for $V_{th} = 1$ –2 mV and at 55 K for 0.3–0.5 mV—one grid step—with depth 0.61–0.78. At 5 mV the criterion lands inside the flux-flow foot [cf. Fig. 1(a)] and the apparent minimum moves to 35 K; this is the documented pathology of large thresholds, not a property of the switching instability.

Descriptor-set sensitivity (Fig. 2(c)). Change point of $PC1(T)$ and its 68% bootstrap interval for each descriptor set: all six, 47.5 K [37.5, 47.5]; (3)–(5) only, 37.5 K [37.5, 37.5]; without (1), 27.5 K [27.5, 32.5]; without (2), 52.5 K [52.5, 52.5]; without (3), 77.5 K [27.5, 77.5]; without (4), 77.5 K [77.5, 77.5]; without (5), 47.5 K [42.5, 47.5]; without (6), 52.5 K [52.5, 52.5]. The spread across sets (27.5–77.5 K) is a systematic of the analysis choice and is much larger than seven of the eight bootstrap intervals (each at most 10 K wide); without (3) the bootstrap itself alternates between the two features of $PC1(T)$. The two 77.5-K values pick up the second feature near T_{BKT} .

Halperin–Nelson systematics (Fig. 5(b)). Across a grid of fit windows (lower edge 85.5–87 K, upper edge 93–97 K) and the four bias currents, T_{BKT} spans 86.4–87.0 K. For the baseline window (86–95 K, $1 \mu\text{A}$): $R_0 = 2.10 \Omega$, $b = 2.50$, $T_{BKT} = 86.92$ K, with parameter correlations $r(b, T_{BKT}) = -0.91$, $r(R_0, b) = +0.99$; residuals show a coherent $\pm(0.02$ – $0.04) \Omega$ structure shared by all four currents. We quote the window spread, not the (much smaller) statistical errors, as the uncertainty.

Synthetic validation of the drift decomposition (Fig. 5(c)). Synthetic datasets use the measured smoothed $I_c(T)$, the measured intrinsic hysteresis, per-scan noise of $0.3 \mu\text{A}$, and an injected inter-file offset $\delta T = 0.3$ K. Over 200 realizations the estimator $\hat{\delta T} = \text{median}[-h_{anti}/(dI_c/dT)]$ recovers 0.30 K, and h_{sym} is unbiased to $< 0.2\%$. Adding an intrinsic polarity asymmetry of 20% of the hysteresis leaves h_{sym} unbiased ($< 0.1\%$) but shifts $\hat{\delta T}$ to 0.99 K: the antisymmetric channel cannot distinguish drift from intrinsic polarity asymmetry, which is why we quote it only as a combined bound.

Observable correlations. Curve-level Pearson correlations over 2–88 K: $r(|dI_c/dT|, 1 - I_r/I_c) = 0.89$; $r(|dI_c/dT|, |\eta|) = 0.35$; $r(|dI_c/dT|, |dPC1/dT|) = 0.40$; $r(1 - I_r/I_c, |\eta|) = 0.48$; $r(1 - I_r/I_c, |dPC1/dT|) = 0.26$; $r(|\eta|, |dPC1/dT|) = 0.27$. In the joint bootstrap the peak locations of $|dI_c/dT|$, $1 - I_r/I_c$, $|\eta|$ and $|dPC1/dT|$ are identical in all 400 resamples (45, 50, 50, 50 K), so their statistical covariance vanishes at the grid resolution and each position carries the grid-limited uncertainty ± 2.5 K.

Appendix C: Thermal power-balance model and RCSJ comparison

Implied thermal conductance. On the falling branch, the last dissipative samples before retrapping give the retrapping power $P = |V||I|$ (averaged over the three samples above I_r). Setting $P = G(T_j - T_{\text{bath}})$ with the hot-spot temperature T_j fixed by $I_r = I_c(T_j)$ —inverting the measured, monotone part of $I_c(T)$ ($T \leq 55$ K, PCHIP interpolation)—yields G at each bath temperature [Fig. 6(a)]. Where the device retraps from the fully resistive branch ($T \leq 15$ K, $V_r = 15\text{--}37$ mV) we find $G = 0.08\text{--}0.13$ $\mu\text{W/K}$ with hot-spot excursions of 10–20 K; where it retraps from the mV-scale flux-flow state (20–50 K) the implied $G \approx 0.01$ $\mu\text{W/K}$ is a lower bound,

since V_r there is limited by the 1-mV criterion.

In-sample reproduction. A smooth two-parameter interpolation $G(T) = G_0 (T/25 \text{ K})^m$ is inserted back into the retrapping condition evaluated on the measured falling branches: descending in $|I|$, the model retrapping current is the first current at which $I_c(T_{\text{bath}} + |V||I|/G) \geq |I|$. The result [Fig. 6(b)] reproduces the measured $I_r/I_c(T)$ including the non-monotonic minimum at 50 K with no further adjustment. Since $G(T)$ was itself obtained from the same retrapping points, this is a consistency check of the power-balance picture, not an out-of-sample prediction; a leave-one-temperature-out calibration would be the natural next test.

RCSJ comparison. The measured dissipative-branch slope at 2–5 K is $R_d = 15.4$ k Ω . In RCSJ the quality factor is $Q = \sqrt{2eI_c R_d^2 C / \hbar}$ and, for $Q \gg 1$, $I_r/I_c \simeq 4/\pi Q$. With $I_c = 64$ μA and a geometric $C = 0.1$ pF (1 pF), $Q \simeq 2.2 \times 10^3$ (6.8×10^3), predicting $I_r/I_c \simeq 6 \times 10^{-4}$ (2×10^{-4})—three orders of magnitude below the measured 0.93. Reproducing 0.93 would require $C \simeq 0.04$ aF, orders of magnitude below any physical junction capacitance. Capacitive hysteresis is therefore excluded quantitatively, and its predicted monotonic temperature dependence is additionally inconsistent with the observed minimum.

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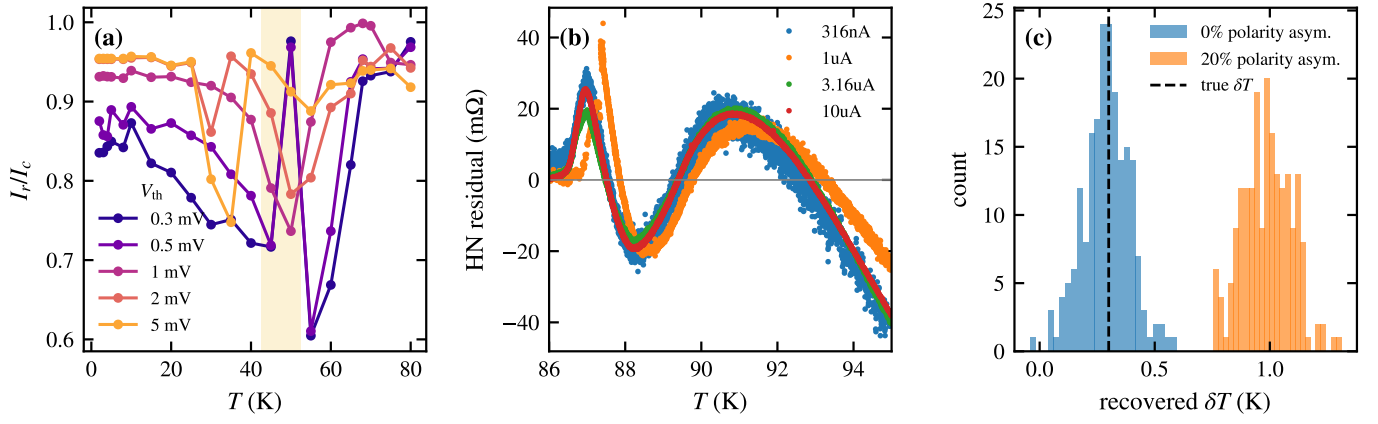


FIG. 5. Robustness tests. (a) $I_r/I_c(T)$ for voltage criteria 0.3–5 mV; the minimum stays at 50–55 K for 0.3–2 mV, while 5 mV enters the flux-flow foot (see text). Shading: T^* band. (b) Halperin–Nelson fit residuals for the four bias currents (baseline window); the shared $\pm(0.02\text{--}0.04) \Omega$ structure sets the systematic scale. (c) Synthetic validation of the drift decomposition: recovered δT distributions without (blue) and with (orange) a 20% intrinsic polarity asymmetry; dashed line marks the injected 0.3 K.

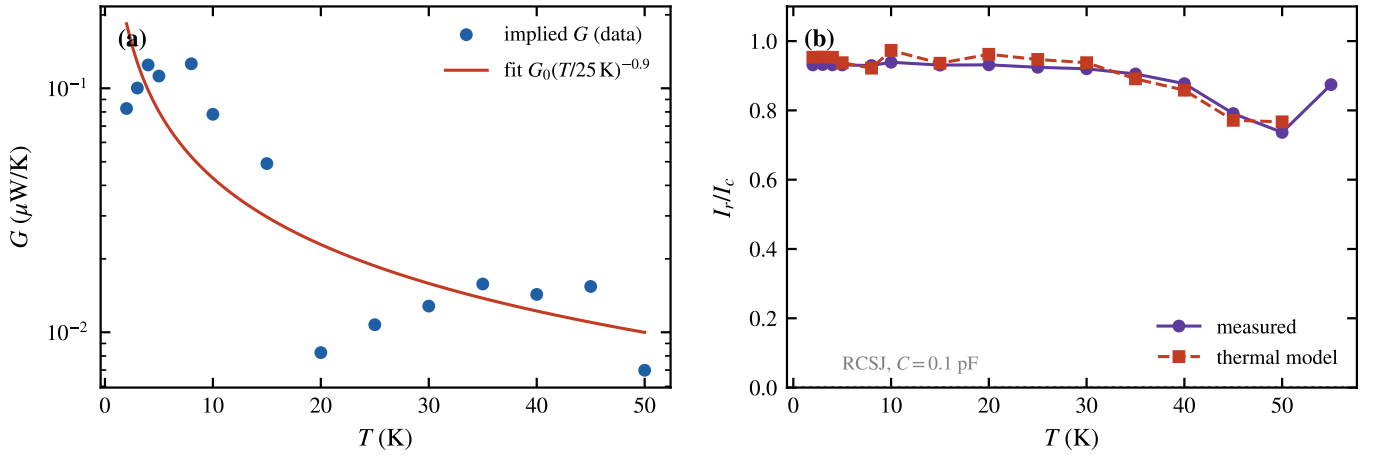


FIG. 6. Thermal power-balance analysis. (a) Effective thermal conductance implied by the measured retrapping points; the line is the smooth interpolation used in (b). Points for $T \geq 20$ K are lower bounds (see text). (b) Measured $I_r/I_c(T)$ versus its in-sample reproduction by the calibrated model; the dotted line is the RCSJ expectation for $C = 0.1$ pF.